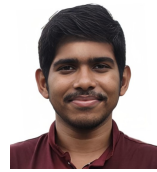


Pranay Kumar Pandey

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Education

Indian Institute of Space Science & Technology, Thiruvananthapuram

M.Tech in Manufacturing Technology, Dept. of Aerospace Engineering

2025 – 2027

Kerala, India

Amity University Mumbai

B.Tech in Aerospace Engineering — CGPA: 8.28

2021 – 2025

Mumbai, Maharashtra

St. Andrews Jr. College, Panvel

Higher Secondary Certificate (HSC), 89.7%

2019 – 2021

Mumbai, Maharashtra

Convent of Jesus & Mary High School, Kharghar

Secondary School Certificate (SSC), 85%

2008 – 2019

Mumbai, Maharashtra

Interest

- Additive Manufacturing
- Finite Element Analysis
- Propulsion Systems
- Composite Manufacturing
- Computer Aided Manufacturing
- Research & Development
- Subtractive Manufacturing
- Fluid Power Systems
- Generative AI

Experience

IIST Thiruvananthapuram

Teaching Assistant

Aug 2025 – Present

- Instructing undergraduate students in CAD Lab Sessions (Sem 1), Manufacturing and Metrology Lab Practices (Sem 4)
- Evaluating lab-based activities of the students

FOSSEE, IIT Bombay

Research Assistant

Mar 2025 – Jul 2025

- Mentoring a cohort of interns and fellows on simulation projects, ensuring quality and technical depth
- Collaborating on national workshops and capacity-building programs to scale CFD outreach
- Expertise in Computational Fluid Dynamics (CFD) and OpenFOAM implementation
- Upgrading Case Studies and Migrating to latest version of OpenFOAM

IIT Kharagpur

Aeroengine Development Intern

Jun 2024 – Jul 2024

- Conducted comprehensive parametric analysis of Variable Cycle Engine performance using GasTurb 14 software
- Performed detailed thermodynamic cycle analysis and evaluated engine performance under various operating conditions
- Evaluated engine performance across take-off, subsonic, and supersonic flight regimes
- Analyzed VABI configurations and their impact on thrust and fuel consumption
- Optimized design parameters including bypass ratios, pressure ratios, and turbine entry temperature

IIT Bombay

FOSSEE Summer Fellowship

May 2023 – Aug 2023

- Conducted numerical study of Non-Newtonian Fluid Flow across Sinusoidal Microchannels using OpenFOAM
- Analyzed velocity profiles, pressure gradients, and viscosity for complex biological and industrial fluids
- Implemented rheological models in open-source and commercial CFD software platforms
- Developed expertise in CFD pre-processing, post-processing, and scientific writing methodologies
- Gained extensive knowledge in literature survey, scientific writing, and pre & post-processing techniques

Skills

- **Programming & Software:** Abaqus, Ansys Fluent, Ansys Workbench, Autocad, C/C++, CNCTRAIN, Fusion 360, GasTurb, Github, ICEM, Java, Mastercam, Matlab, Openfoam, Openvsp, Paraview, Python, Scilab, Simufact Additive, Simulink, SolidWorks, Ultimaker Cura
- **Soft Skills:** Leadership, Effective Communication, Public Speaking
- **Languages:** Hindi, Marathi, English, German (A2)

Achievements and Awards

- Emerging Talent in Open Source Software - National FOSSEE Awards 2025
- Qualified GATE AEROSPACE - 2025, AIR 651
- Global Qualifier, NASA Space Hackathon (Oct 2024)
- First Prize, CFD Hackathon by FOSSEE, IIT Bombay (Dec 2023)
- NPTEL Course Topper, Aeroengine Gas Turbine Cycle, IIT Kharagpur (Oct 2023)
- FOSSEE Summer Fellowship, IIT Bombay (May 2023)

Conferences

- System Dynamics Theory & Practice, 2024 (IEOR, IIT Bombay)
- International Conference on Advances in Aerospace & Energy Systems (IAES), 2024 (ISRO LPSC - IIST, Kerala)

Projects Undertaken

- **Development of Random Positioning Machine for Simulating Microgravity on Earth (M.Tech Course Project, 2025):** Working on Model Development and Digital Twin
- **Advancement in Affordable and Accessible Avionics Control Systems for Modern Rocketry (B.Tech Project, 2025):** Developed cost-effective, robust avionics for model rocketry, focused on lightweight structures and reliability.
- **Autonomous Drone Navigation in GPS-Denied Martian Terrain (IROC-U, 2024):** Systems Engineer managing Sensor Fusion, Script Development and adherence to Competition Outcomes.
- **Conceptual Supersonic UAV Design for Military Applications (NACDEC VIII, 2024):** Team Mentor : Contributions to aerodynamic and propulsion system design for supersonic flight.

References

- **Dr. Sooraj V** - Professor, Department of Aerospace Engineering, IIST - Trivandrum — sooraj@iist.ac.in
- **Dr. Tushar Chourushi** - Associate Professor, Aerospace Engineering, MIT-ADT University, Pune — tushar.chourushi@mituniversity.edu.in
- **Dr. Swagata Paul** - Assistant Professor, Department of Aerospace Engineering, Ajeenkya DY Patil University, Pune — pswagata82@yahoo.com